

# Ethan Wayne Holbrook

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West Lafayette, IN, 47906

## EDUCATION

### Doctor of Philosophy – Materials Engineering

2021 – 2026

Computational Engineering Concentration || *Purdue University*, West Lafayette, IN

### Bachelor of Science – Materials Science & Engineering

2017 – 2021

Mathematics Minor || *Wright State University*, Dayton, OH

## PROFESSIONAL EXPERIENCE

### Graduate Research Assistant

2021 – Present

*Purdue University*, West Lafayette, IN

- Led atomistic simulations of shock-driven jetting in multi-component high explosives, showing how shock strength, crystallography, and interfacial structure govern jet formation, mixing, and localized heating relevant to hotspot formation.
- Built a parsing and validation framework for simulation inputs, including systematic evaluation of LLM-generated scripts to identify failure modes and support more autonomous molecular dynamics workflows.
- Investigated high-strain-rate deformation in aluminum single crystals using molecular dynamics and OVITO, characterizing defect evolution, plasticity, and microstructural response under extreme loading.

### LLNL Intern (CCMS/MaCI)

May-Aug. 2022 – 2024

*Lawrence Livermore National Laboratory*, Livermore, CA

- Performed force-field validation using equation-of-state and vibrational spectra / normal mode analyses, helping establish fidelity across thermodynamic and molecular-response metrics.
- Combined validated TNT and HMX force fields into a unified framework for atomistic simulation of Octol, enabling molecular-level investigation of multi-component energetic materials.
- Developed computational workflows in LAMMPS for constructing arbitrary molecular-crystal interfaces.

### AFRL RXAP SOCHE Intern

2020 – 2021

*Air Force Research Laboratory*, Dayton, OH

- Supported setup of a laser-heated pedestal growth lab by organizing equipment and preparing spaces for research use and developing custom lab components in SolidWorks.
- Measured thin-film optical properties using spectroscopic ellipsometry and refined YAG fiber polishing procedures to meet analysis requirements.

## PUBLICATIONS

### Peer-Reviewed Articles

- [1] Ethan Holbrook, Matthew P. Kroonblawd, Brenden W. Hamilton, H. Keo Springer, and Alejandro Strachan. "Modeling Framework to Predict Melting Dynamics at Microstructural Defects in TNT-HMX High Explosive Composites". In: *Journal of Physical Chemistry C* 129.31 (2025), pp. 14111–14129. ISSN: 19327455. DOI: 10.1021/acs.jpcc.5c02261. arXiv: 2503.23680.
- [2] Chunyu Li, Saswat Mishra, Ethan Holbrook, Chukwuma Ezenwata, and Alejandro Strachan. "High strain rate deformation mechanisms in fcc alloys as a function of load triaxiality". In: *Phys. Rev. Mater.* 9 (11 2025), p. 113604. DOI: 10.1103/1jzt-x6k4. URL: <https://link.aps.org/doi/10.1103/1jzt-x6k4>.

### Preprints

- [1] Juan Carlos Verduzco, Ethan Holbrook, and Alejandro Strachan. "GPT-4 as an interface between researchers and computational software: improving usability and reproducibility". In: *arXiv preprint arXiv:2310.11458* (2023).

### Submitted

- [1] Juan Carlos Verduzco and Ethan Holbrook. "Scientific Responsible Autonomy in Agentic AI". In: *Modelling and Simulation in Materials Science* (2025).

### In Preparation

- [1] Ethan Holbrook, Matthew P. Kroonblawd, H. Keo Springer, and Alejandro Strachan. "Shock-driven Jetting in HMX: Part 1 – Jetting". In: (2026).
- [2] Ethan Holbrook, Matthew P. Kroonblawd, H. Keo Springer, and Alejandro Strachan. "Shock-driven Jetting in HMX: Part 2 - Jet Impact & Mixing". In: (2026).
- [3] Ethan Holbrook, Juan Carlos Verduzco, and Alejandro Strachan. "Integrating AI into Molecular Dynamics Workflows: LLMs and Static Parsing for LAMMPS Script Automation". In: (2026).

## Conference Presentations and Proceedings

- [1] Ethan Holbrook, Matthew Kroonblawd, Keo Springer, and Alejandro Strachan. "Molecular Simulations of Shock Interactions with Microstructural Defects in a Two-Component High Explosive System". In: *SCCM 2025*. 2025.
- [2] Ethan Holbrook, Juan Carlos Verduzco, Kat Nykiel, William Zummo, and Alejandro Strachan. "AI-Powered Interface: Fully Automated Tool for LAMMPS Simulation and Analysis". In: *TMS2025 Technical Program*. 2025.
- [3] Ethan Holbrook, Juan Carlos Verduzco, and Alejandro Strachan. "Natural Language to LAMMPS: LLMs as interfaces between researchers and scientific software". In: *LAMMPS Workshop and Symposium*. 2025.
- [4] Juan Carlos Verduzco, Ethan Holbrook, and Alejandro Strachan. "Teaching AI to Speak LAMMPS: A Practical Guide to Prompting and Script Checking". In: *LAMMPS Workshop and Symposium*. 2025.
- [5] Ethan Holbrook, Matthew Kroonblawd, Keo Springer, and Alejandro Strachan. "Interfacial Properties of Heterogeneous Energetic Materials: A Molecular Dynamics Study". In: *APS March 2024*. 2024.
- [6] Ethan Holbrook, Matthew Kroonblawd, Keo Springer, and Alejandro Strachan. "Molecular Insights into Interface Dependency for TNT Melting". In: *GRC 2024 Poster*. 2024.
- [7] Ethan Holbrook, Juan Carlos Verduzco, and Alejandro Strachan. "Simplifying Computational Simulations: Using Large Language Models for Automated Research in Materials Science". In: *Hands-on Data Science and Machine Learning Training Series Webinar*. 2024.
- [8] Ethan Holbrook, Matthew Kroonblawd, Keo Springer, and Alejandro Strachan. "HMX/TNT Interfacial Properties from Molecular Dynamics Simulations: Energetics, Rheology, and Molecular Structure". In: *APS SCCM 2023*. 2023.
- [9] Ethan Holbrook, Michael Cowan, and Giannis Mpourmpakis. "Elucidating Structure-Property Relationships of Atomically-Precise Gold Nanoclusters from Density Functional Theory". In: *AIChE 2019*. 2019.

## HONORS AND AWARDS

### Summa Cum Laude

Wright State University

2021

### Valedictorian Scholarship

Wright State University

2017

## TEACHING EXPERIENCE

### Graduate Mentor - Undergraduate Research Programs

Purdue University

May – Aug. 2025

- Mentored Purdue undergraduate students on research projects combining material science, data science, and automated FAIR workflows.

### Graduate Teaching Assistant - Structure and Properties of Materials (MSE 230)

Purdue University

Spring 2023

- Led recitation sessions for an introductory materials science course with 40 undergraduate students, giving lectures on fundamental materials science concepts.
- Developed exam-review content and fielded general materials science questions, providing individualized assistance to support learning across knowledge levels.

### Undergraduate Teaching Assistant - Circuit Analysis I

Wright State University

Fall 2019

- Facilitated classroom activities and provided live instructional support for 100+ students

## RELEVANT SKILLS

### Programming Languages

- Proficient: Python, LaTeX
- Familiar: Bash, Fortran90, MATLAB
- Tools: HPC, SLURM, Git

### Machine Learning and Data Science

- Frameworks: TensorFlow/Keras, PyTorch
- Libraries: Pandas, Scikit-learn, Plotly

### Software

- Computational Packages: LAMMPS, Quantum ESPRESSO
- Analysis: OVITO, Avogadro
- Design: SolidWorks
- Others: LabVIEW, Microsoft Office Suite

## PROFESSIONAL ACTIVITIES

### Workshop Organization

- nanoHUB's "AI 4 Science Challenge" competition (2024), Purdue University — Organizer.
- MaRDA Annual Meeting (2024), Virtual — Poster session coordinator.

**STUDENT MENTORING**

| <b>Name</b>            | <b>Supervisor</b>      | <b>Student Status</b> | <b>Project Topic</b>                      | <b>Year</b> |
|------------------------|------------------------|-----------------------|-------------------------------------------|-------------|
| Pablo Juanes Gutierrez | Dr. Alejandro Strachan | Undergraduate         | Molecular Crystal Surface Energy Database | 2025        |